Assessment of drought in northern latur district (2010-2022): an integrated approach using remote sensing and meteorological data

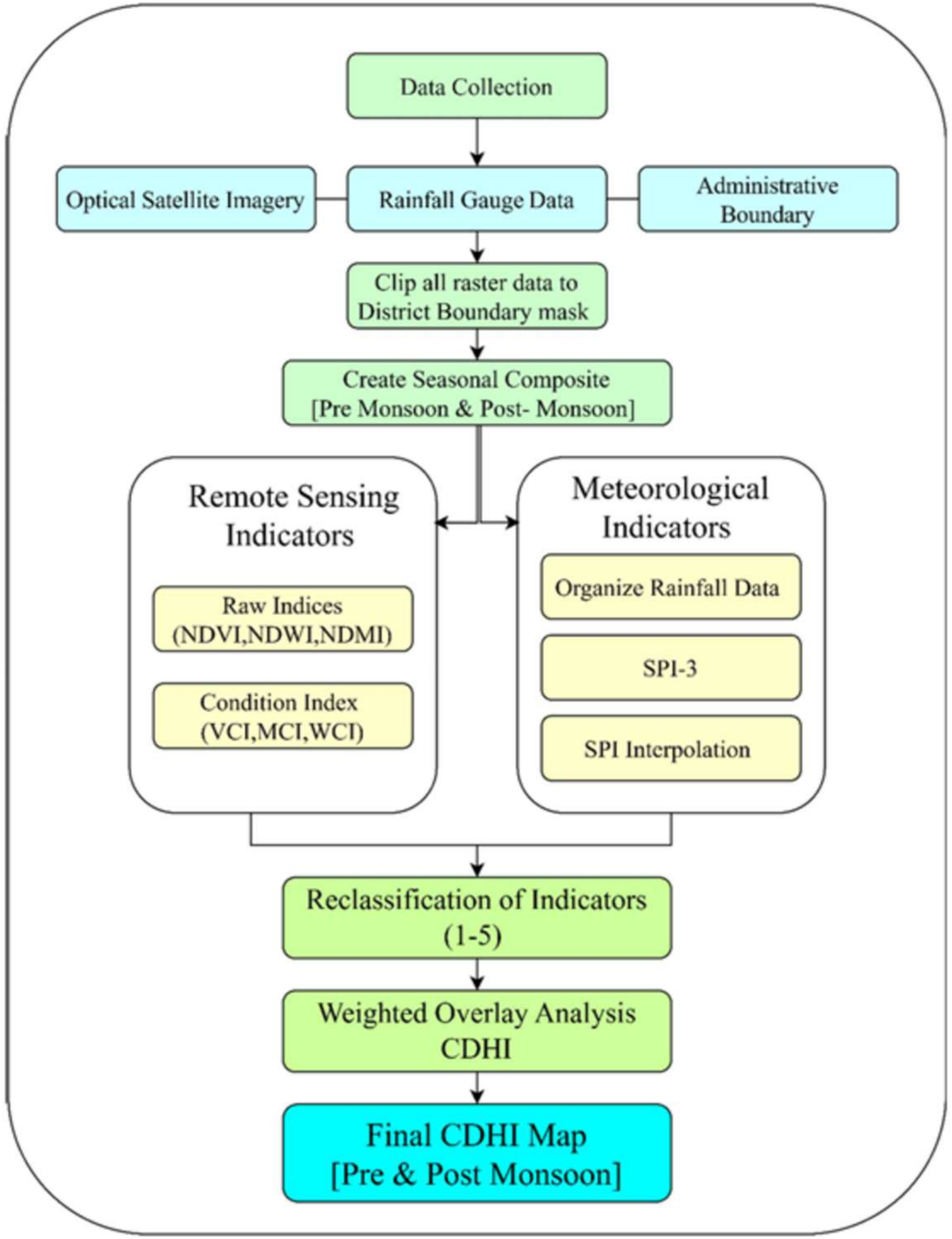
Introduction of Project :-

Drought in India is a multifaceted crisis, particularly in the agricultural heartlands of Maharashtra. In the Marathwada region, the Northern Latur District serves as a stark case study for climate vulnerability, where the economy’s reliance on variable monsoon rains has led to historic water crises—most notably the 2016 "Jaldoot" water train emergency. While rainfall deficits are the primary trigger, traditional monitoring often fails because it treats drought as a single-dimensional event. In reality, a meteorological deficit cascades into agricultural and hydrological stress, impacting soil moisture and reservoir levels at different scales. This project addresses the critical lack of integrated monitoring by developing a Composite Drought Hazard Index (CDHI). By synthesizing meteorological data with spatially explicit satellite remote sensing indices, this research provides a holistic framework to quantify drought severity, offering a more nuanced and scientific approach to disaster mitigation and resource management in one of India’s most chronic drought hotspots.

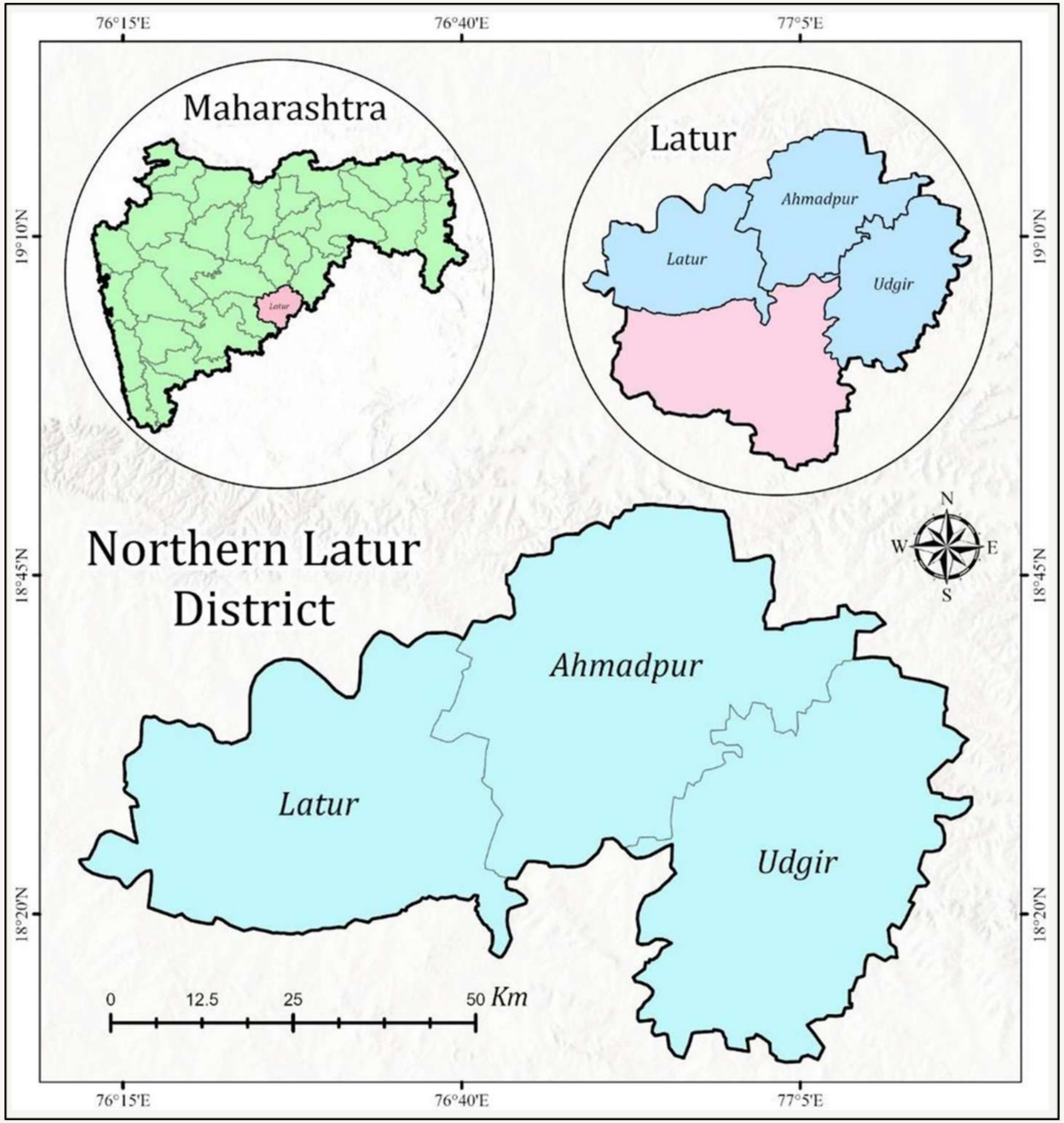
Data Sours :-

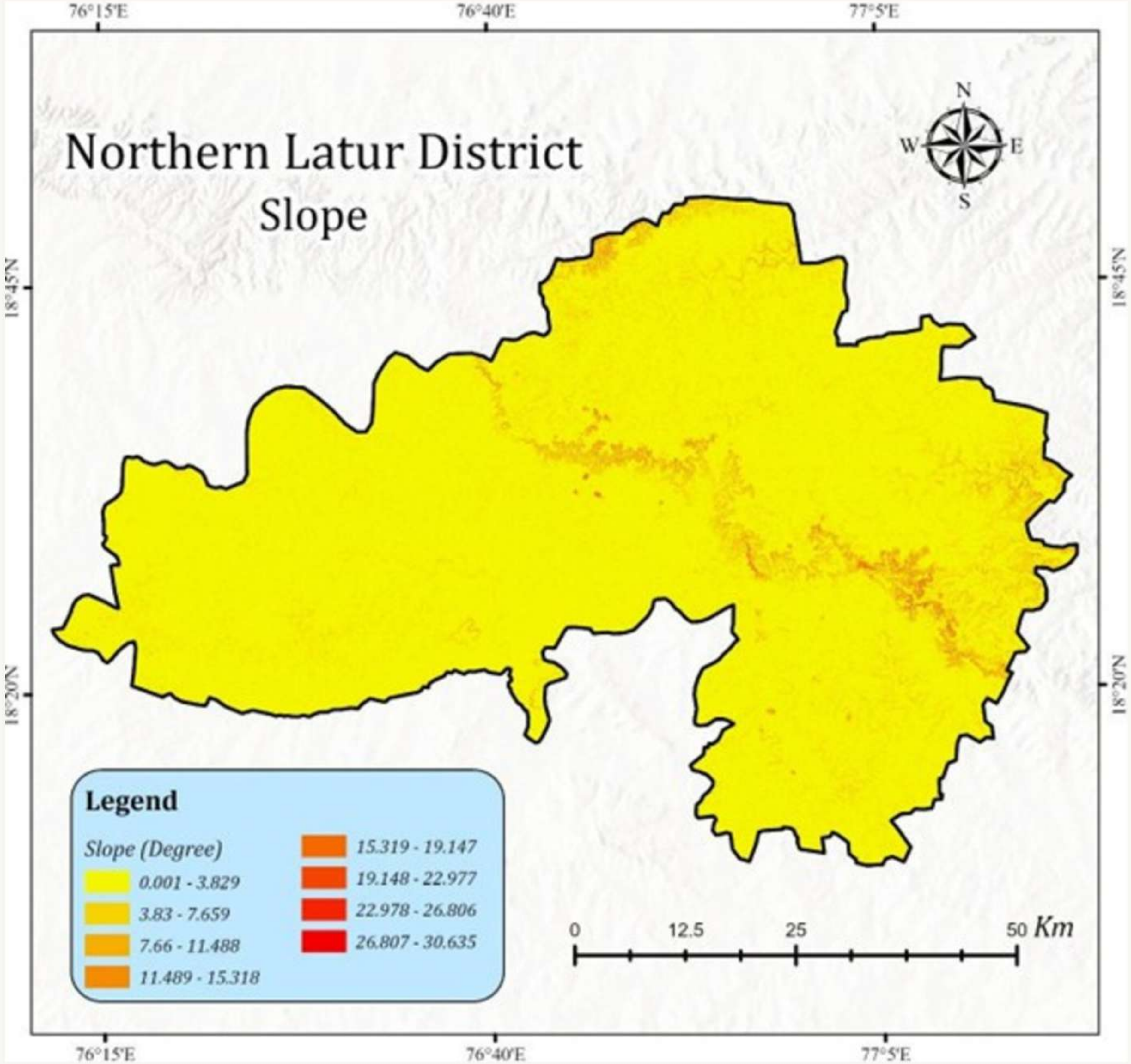
Data Type	Source/Platform	Sensor/Station Network	Spatial Resolution	Period of Record Used	Key Parameters Derived
Optical Satellite Imagery	USGS Earth Explorer	Landsat 5 TM / 7 ETM+ / 8 OLI/TIRS	30 m	2010-2022	NDVI, NDWI, NDMI
Rainfall Gauge Data	IMD DatasetS	Latur District Stations	Point Data	2010-2022	Monthly Precipitation
Administrative Boundary	Survey of India	N/A	Vector Polygon	N/A	Study Area Mask

Methodology Flow Chart:

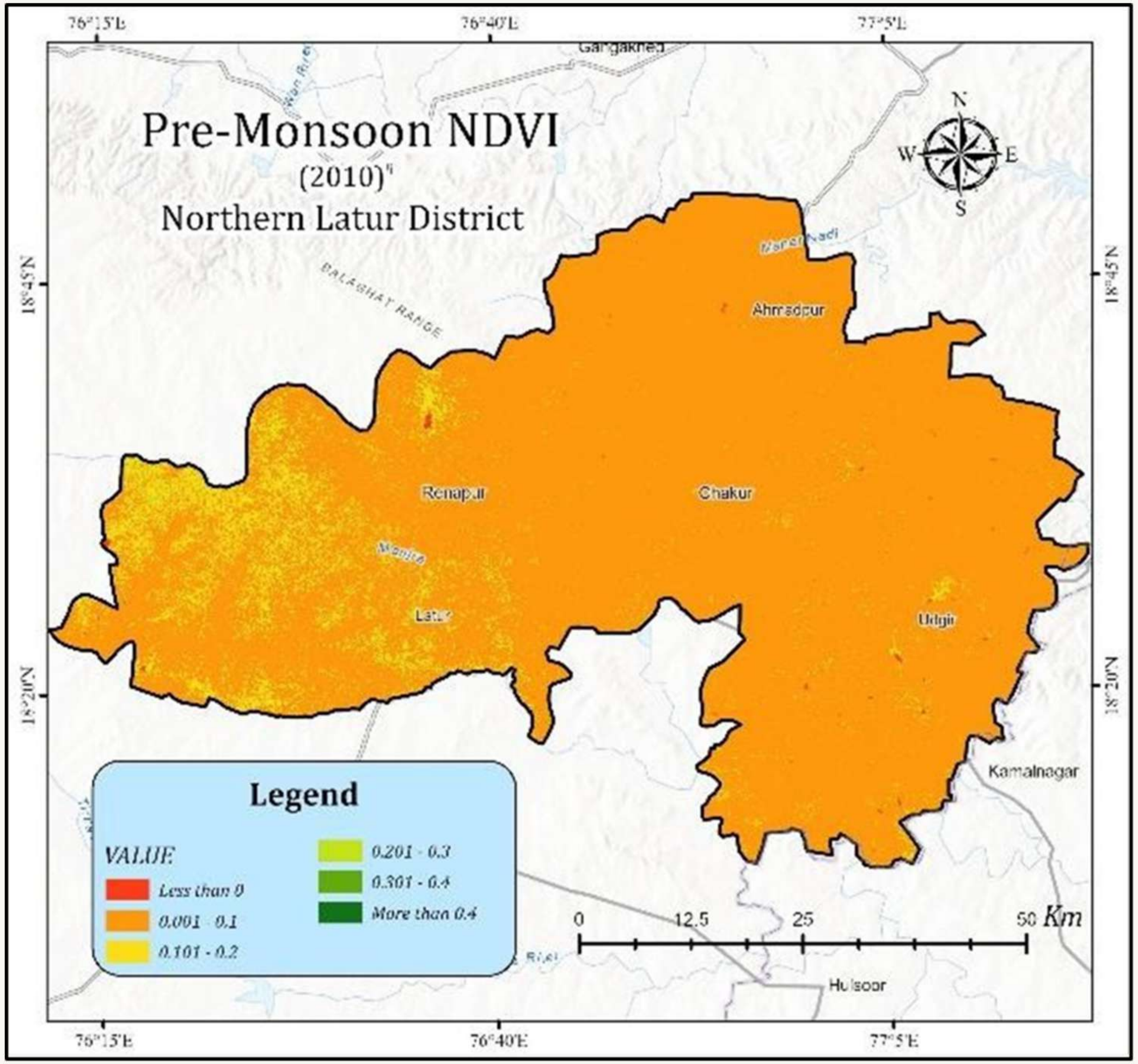


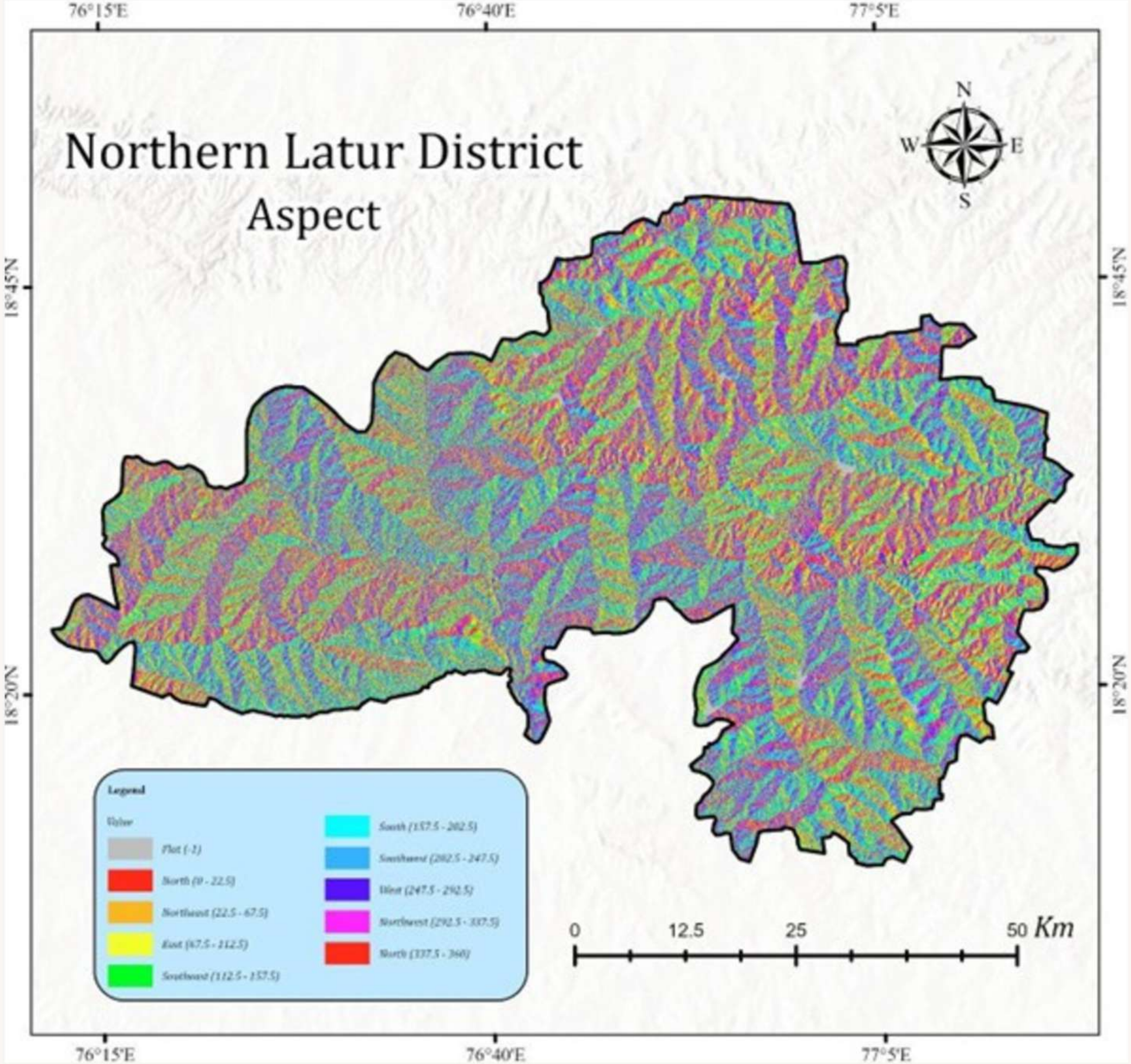
Study area map :-



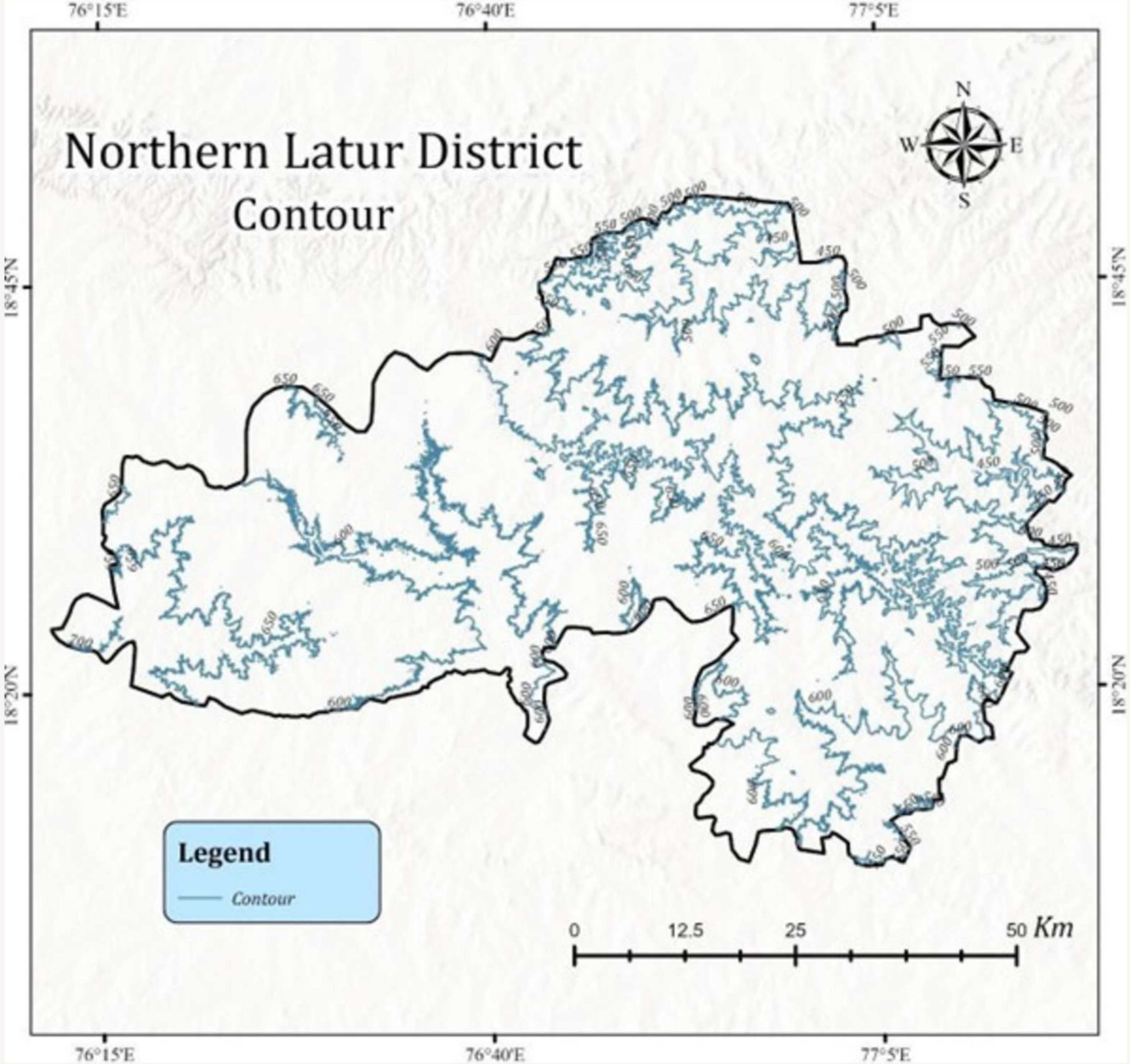


76°15'E
76°40'E
77°5'E
18°45'N
18°20'N
Northern Latur District
Slope
N
W
E
S
Legend
Slope (Degree)
0.001 - 3.829
3.83 - 7.659
7.66 - 11.488
11.489 - 15.318
15.319 - 19.147
19.148 - 22.977
22.978 - 26.806
26.807 - 30.635
0
12.5
25
50 Km


Pre-Monsoon NDVI
(2010)
Northern Latur District
76°15'E
76°40'E
77°5'E
18°45'N
18°20'N
N
S
E
W
Gangakhed
Balaghat Range
Ahmadpur
Renapur
Chakur
Latur
Udgir
Kamalnagar
Hulsoor
Manjra
Legend
VALUE
Less than 0
0.001 - 0.1
0.101 - 0.2
0.201 - 0.3
0.301 - 0.4
More than 0.4
0
12.5
25
50 Km



Northern Latur District
Aspect
N
W
E
S
76°15'E
76°40'E
77°5'E
18°45'N
18°20'N
Legend
Value
Flat (-1)
North (0 - 22.5)
Northeast (22.5 - 67.5)
East (67.5 - 112.5)
Southeast (112.5 - 157.5)
South (157.5 - 202.5)
Southwest (202.5 - 247.5)
West (247.5 - 292.5)
Northwest (292.5 - 337.5)
North (337.5 - 360)
0
12.5
25
50 Km


Northern Latur District
Contour
Legend
Contour
0
12.5
25
50 Km
N
S
E
W
76°15'E
76°40'E
77°5'E
18°45'N
18°20'N